

Multi-Hop Retrieval-Augmented Generation through the Lens of Evidence Chains:

A Diagnostic Survey

YUQING LUO, University of Science and Technology of China, China

KAI ZHANG, University of Science and Technology of China, China

Multi-hop retrieval-augmented generation (RAG) serves information needs whose support is distributed across evidence units and whose later retrieval steps depend on intermediate results. This dependency creates an information retrieval problem beyond ranking several independently relevant documents: a system must acquire at least one valid evidence chain, preserve it under a passage, token, or retrieval-call budget, organize it for reader access, compose its elements correctly, and produce an answer that is faithfully supported by the supplied evidence. Existing surveys mainly organize the literature by architecture or control flow, which obscures the stage at which a method acts and the evidence needed to validate its claim. We present an IR-centered diagnostic survey organized around five evidence-chain targets: chain observability, budgeted chain preservation, evidence exposure, cross-evidence fusion, and causal faithfulness. We synthesize sparse, dense, iterative, graph-guided, hierarchical, and agentic retrieval; bridge- and path-aware selection; compression and context organization; fusion and decomposition methods; and verification and attribution approaches through July 2026. Across these families, we compare the state used to issue the next query, the unit of selection, the operative budget, the supervision signal, the reported diagnostic, and the characteristic failure mode. We then map benchmark annotations and metrics to the targets they can and cannot identify, distinguish grounded success from ordinary answer accuracy, and propose a conditional reporting protocol covering retrieval depth, context budget, membership, ordering, reader capability, faithfulness, and cost. The framework is intended as a comparative and evaluation tool rather than a predictive theory of system performance. A versioned companion artifact records the review protocol, coded entries, and provenance of taxonomy assignments.

Additional Key Words and Phrases: multi-hop retrieval, retrieval-augmented generation, evidence chains, evidence selection, context organization, question answering, faithfulness, evaluation

1 INTRODUCTION

Retrieval-augmented generation is often described as a two-stage system: retrieve documents and generate an answer. That description is inadequate for questions whose evidence is distributed across sources and connected by dependencies. In a multi-hop question, an answer-bearing passage may not be retrievable from the original query at all. It becomes identifiable only after an earlier passage reveals a bridge entity, relation, date, or partial conclusion. The retrieval problem is therefore sequential and conditional: the system must discover a support structure, not merely collect several high-scoring documents.

This distinction matters for both system design and evaluation. A complete support chain may be absent from the initial retrieval pool. It may be present in the pool but removed by a top- K selector that favors marginal query relevance. It may survive selection but be placed in a reader-unfriendly position, separated from its companion evidence by distractors, or compressed in a way that removes the bridge relation. Even when all required evidence is available, the reader may fail to combine it. Conversely, a correct answer may be produced from parametric memory, annotation artifacts, or a single-hop shortcut without using the retrieved chain. End-to-end exact match or F1 collapses these cases into one number and therefore gives limited guidance about where a method succeeds.

Authors' addresses: Yuqing Luo, University of Science and Technology of China, Hefei, Anhui, China; Kai Zhang, University of Science and Technology of China, Hefei, Anhui, China.

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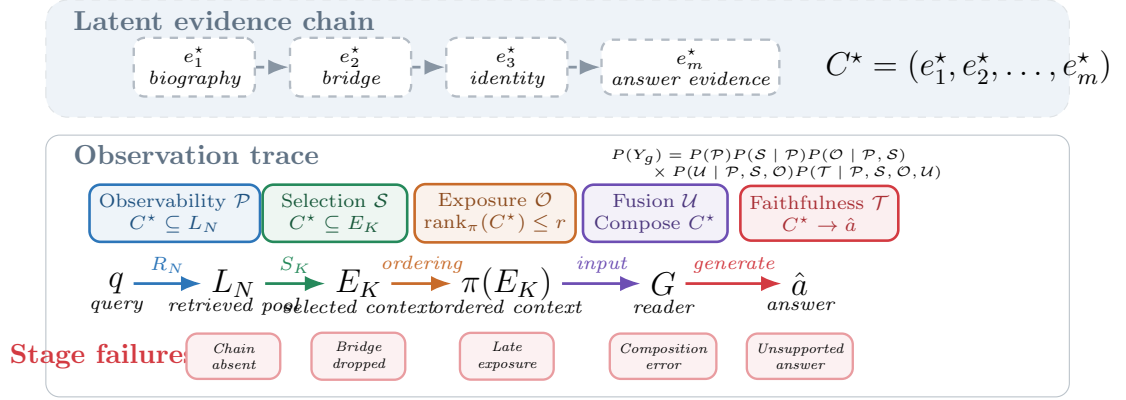


Fig. 1. An evidence-chain view of multi-hop RAG. A system constructs a retrieval pool L_N , selects a budgeted context E_K , presents the context through an ordering or packing policy π , and generates an answer. Evaluation should separately test whether a valid chain is observable, preserved, exposed, correctly fused, and causally used. Ordinary answer accuracy also includes answers obtained through shortcuts, parametric memory, or guessing.

For a question q , let $\mathfrak{C}^*(q)$ denote the set of valid minimal evidence chains that support the reference answer. A retrieval system constructs a candidate pool L_N , a selector places a subset E_K under a passage or token budget, an ordering or packing policy produces $\pi(E_K)$, and a reader generates $\hat{a}$. We use this sequence to organize the literature around five diagnostic questions:

- (1) Does L_N contain at least one valid chain?
- (2) Conditional on chain presence, does the selected context preserve one?
- (3) Conditional on preservation, is the chain presented where and how the reader can access it?
- (4) Given accessible evidence, can the reader perform the required cross-evidence composition?
- (5) Is the resulting answer causally supported by the supplied evidence?

These questions correspond to chain observability, selection preservation, evidence exposure, fusion reliability, and causal faithfulness. They are not independent modules, and a single system change can affect several of them. Their value is diagnostic: they separate claims that are routinely conflated in multi-hop RAG experiments.

The product shown in Figure 1 is the exact chain-rule decomposition of the grounded-success event Y_g under the declared evaluation protocol; it is not an approximation to ordinary answer accuracy.

Existing surveys provide broad taxonomies of RAG architectures, multi-hop question answering, graph reasoning, chain-of-thought, and agentic retrieval [21, 30, 72, 85, 91, 101, 117]. These taxonomies answer an important procedural question: how is a system built? They are less suited to a second question: which failure event does the central mechanism change? An iterative dense retriever, a graph expander, and a search agent have different architectures, yet all can increase the probability that a bridge-dependent chain enters the candidate pool. A cross-encoder, a bridge-aware selector, and a compressor can all change which chain elements survive the final budget. A long-context model and an ordering policy can both affect evidence access without altering retrieval recall.

The survey therefore uses two complementary views. The primary view follows the evidence chain from acquisition to causal use. A compact architectural crosswalk is retained to connect this organization to familiar method families. This positioning is deliberately narrower than a general RAG or LLM-reasoning survey. We focus on systems in

Table 1. Positioning of the present survey relative to major survey families.

Survey family	Representative scope	Primary organizing axis	Gap addressed here
General RAG [21, 30]	Retrieval, augmentation, and generation	System architecture and lifecycle	Multi-hop failure stages are usually aggregated.
Multi-hop QA / MRC [85, 91]	Bridge, comparison, and compositional reasoning	Task and reasoning type	Pool, budget, presentation, and evidence use are not separated.
Reasoning and agentic RAG [72, 117]	Adaptive retrieval, planning, and tool use	Control flow and agent design	End-to-end gains remain difficult to attribute.
Retrieval-reasoning process [49]	Plan, index, control, and stopping	Procedural decisions	No stage-conditional metric and reporting map.
This survey	Multi-hop RAG over text, graphs, tables, and hybrid sources	Evidence-chain stages plus architecture	Conditional diagnostics, failure attribution, and reporting protocol.

which external evidence and cross-evidence dependency are necessary parts of the task. Purely parametric reasoning, general-purpose agents without retrieval-grounded evidence use, and single-hop retrieval methods are discussed only when they provide a component or baseline for multi-hop systems.

1.1 Position Relative to Existing Surveys

Prior work surveys general RAG [21, 30, 43, 66, 174], multi-hop question answering and reading comprehension [85, 91], complex and knowledge-graph QA [32, 61, 99], step-by-step reasoning [16, 101, 104], attribution and hallucination [42, 64], and agentic or reasoning-oriented RAG [72, 117]. A close process-centric framework organizes retrieval-reasoning systems by execution plan, index structure, next-step control, and stopping criterion [49]. These views explain what systems do. Our complementary view asks which evidence-chain failure stage a mechanism changes and which conditional evidence supports that claim.

The ingredients of this framing are not new in isolation. Retrieved documents have been modeled as latent variables in end-to-end RAG [36, 63]; multi-hop datasets annotate support chains [41, 133, 161]; and attribution work studies whether answers depend on evidence [29, 74, 147, 157]. The contribution is their integration into one IR-centered diagnostic organization that supports cross-architecture comparison and exposes what common metrics do not identify.

The survey is guided by four research questions:

RQ1: Evidence acquisition.

Which retrieval mechanisms make dependent support chains observable, and how do they trade coverage against noise, cost, and query drift?

RQ2: Budget and presentation.

How do selection, compression, ordering, and context construction determine whether an observable chain remains usable by the reader?

RQ3: Composition and support.

Which reader, decomposition, graph, and verification mechanisms improve cross-evidence reasoning and evidence-grounded answers?

RQ4: Evaluation.

Which benchmarks, annotations, and interventions can identify each failure stage, and which common metrics conflate them?

1.2 Contributions

This survey makes four contributions.

- **An IR-centered evidence-chain framework.** We formalize the observable sequence from retrieval pool to generated answer and define five protocol-bound events that make retrieval, selection, presentation, reasoning, and faithfulness claims comparable across architectures.
- **A dual-axis review.** We organize methods primarily by the stage they target while retaining an architectural crosswalk covering iterative retrieval, graph-based methods, decomposition, fusion readers, long-context systems, verification, and agentic search.
- **Benchmark and metric alignment.** We reinterpret supporting-fact, path-annotated, long-context, heterogeneous-evidence, and counterfactual benchmarks according to the failure events they expose, and we distinguish pool recall, context preservation, fixed-membership presentation, oracle-evidence reasoning, and causal-support diagnostics.
- **A reporting protocol and auditable artifact.** We propose a minimum set of conditional diagnostics and budget measures for multi-hop RAG studies, and release the literature protocol and coded records with explicit verification status.

Organization. Section 2 describes the review protocol and scope. Sections 3 and 4 introduce the task model and diagnostic decomposition. Section 5 reviews method families along the evidence-chain pipeline, and Section 6 provides the architectural crosswalk. Section 7 aligns benchmarks and metrics with diagnostic claims. Section 8 focuses on causal evidence use. Sections 9 and 10 summarize open problems and reporting recommendations.

2 REVIEW PROTOCOL AND LITERATURE COVERAGE

The review is designed as a structured evidence map rather than a statistical meta-analysis. Its purpose is to compare mechanisms and evaluation claims across communities whose terminology differs, while preserving a traceable record of why each work is included and how it is coded.

2.1 Search Sources and Time Window

We searched ACL Anthology, OpenReview, arXiv, ACM and other publisher proceedings pages, and official project repositories. DBLP and Semantic Scholar were used for secondary discovery and deduplication. The review covers work from January 2017 through July 2026. Query families combined task terms (e.g., *multi-hop question answering*, *multi-step retrieval*, *compositional question answering*), mechanism terms (e.g., *iterative retrieval*, *graph retrieval*, *question decomposition*, *context compression*, *agentic search*), and evaluation terms (e.g., *supporting facts*, *path recall*, *citation faithfulness*, *counterfactual evaluation*). Backward and forward citation tracing from major surveys, benchmarks, and canonical methods supplemented keyword search.

2.2 Inclusion, Exclusion, and Coding

A work is included when it satisfies at least one of four conditions: it proposes a method for retrieving or using dependent evidence; introduces a benchmark or annotation scheme that exposes multi-step evidence dependency; analyzes evidence use or failure in a retrieval-grounded setting; or provides a foundational component repeatedly used by multi-hop RAG systems. We exclude works whose multi-hop framing is incidental, general agents without a retrieval-grounded evidence objective, and single-hop methods that do not materially inform a multi-hop stage.

Each method record stores its evidence substrate, retrieval setting, context budget, architectural family, primary diagnostic target, secondary targets, benchmarks, cost information, and source-verification status. The primary target is the earliest evidence-chain stage directly changed by the central mechanism; later effects are recorded as secondary

Table 2. Frozen seed audit over source-verified method records. Counts use the reviewed denominator $n = 20$ and do not include discovery-only entries.

Audit item	Count	Interpretation
Primary target: observability	7/20	Acquisition remains the dominant coded intervention target.
Primary target: budgeted selection	3/20	Selection-preservation claims are present but less often isolated than acquisition claims.
Primary target: evidence exposure	2/20	Context organization appears mainly in hierarchical or graph-summary systems.
Primary target: fusion reliability	5/20	Reader and graph-reasoning methods remain a substantial family.
Primary target: causal faithfulness	0/20	No reviewed seed method is primarily coded as a causal-use intervention.
Joint or adaptive primary control	3/20	Agentic systems often span several events and require event-specific logs before attribution.
Public code URL recorded	13/20	Reproducibility metadata is common but not universal in the reviewed seed.
Benchmark records reviewed	6	Most benchmark entries identify joint answer or support behavior rather than a single controlled stage.

targets. For example, a query-rewriting method that expands the candidate pool is coded primarily as observability, even when the resulting evidence also improves fusion.

The coding scheme separates three objects that should not be conflated: *discovery records*, which maximize recall during literature search; *reviewed method and benchmark records*, whose metadata and assignments have been checked against the paper or official project page; and *manuscript citations*, which additionally include background and component papers. Only reviewed records support taxonomy-level comparisons. The companion artifact reports these layers and their status independently.

2.3 Quality Control and Limitations

Borderline assignments are logged with the competing labels, the relevant source passage, and the reason for the final decision. Records whose primary intervention stage remains unresolved after full-text checking stay in a *needs-review* state and are excluded from taxonomy-level comparisons. Because most records are single-coded rather than independently double-coded, we do not report an inter-annotator agreement coefficient. Absence from the catalog is likewise not interpreted as evidence of irrelevance or low quality.

Three limitations follow. First, rapid publication in 2025–2026 means that titles, venue status, and versions can change; the artifact therefore records a cut-off date and canonical source. Second, valid support chains are often non-unique, so benchmark annotations identify only a subset of the chains a system might use. Third, bibliography breadth is not equivalent to verified taxonomy coverage. The companion artifact therefore records a versioned release, the reviewed-record denominator used by generated comparisons, and the common source-level checklist required for every included entry.

2.4 Quantitative Reporting Audit

To avoid treating bibliography size as empirical coverage, we conducted a small reporting audit on the frozen reviewed seed used for this revision. The denominator is the set of method records in the companion artifact with source-verified status, excluding discovery-only and needs-review entries. The audit is therefore a diagnostic sample of the catalog, not a frequency estimate over all multi-hop RAG publications.

Two patterns are visible even in this conservative seed. First, the catalog is front-loaded toward acquisition and fusion: observability and fusion account for 12 of the 20 reviewed method records, whereas selection and exposure are usually evaluated through downstream answer gains rather than conditional chain-preservation or fixed-membership tests. Second, faithfulness is mostly a secondary or evaluation-side concern in the seed. This supports the reporting protocol in Section 10: future catalog releases should code whether each paper reports stage-aligned metrics, token

budget, retrieval-call budget, latency, membership/order controls, and deletion, conflict, or counterfactual evidence-use probes. Those fields are required before stronger temporal or architecture-family reporting-gap claims can be made.

3 PROBLEM SETUP: MULTI-HOP RETRIEVAL AS EVIDENCE-CHAIN ACQUISITION

3.1 What Makes Retrieval Multi-Hop?

A retrieval task is multi-hop when at least one required evidence unit becomes retrievable or interpretable only after an intermediate state has been resolved. This definition emphasizes dependency rather than the raw number of documents. A question can require many passages but remain single-hop if all passages are independently query-relevant. Conversely, a two-document bridge question is multi-hop when the second document cannot be specified before the first reveals an entity or relation.

Three properties are useful in practice. *Semantic non-locality* means that some required evidence is weakly related to the original query surface. *Dependency* means that an intermediate result changes the next retrieval or interpretation step. *Joint sufficiency* means that no proper subset of the selected support is sufficient under the task protocol. These properties distinguish multi-hop RAG from long-document reading, ordinary multi-document aggregation, and difficult single-evidence reasoning.

3.2 Evidence Chains and Observable System Objects

Let q be a question, a^* its reference answer, and $\mathcal{K}$ an external knowledge store. An evidence unit may be a sentence, passage, table row, image region, knowledge-graph triple, or another retrievable object. Let $\mathfrak{C}^*(q)$ denote the set of valid minimal evidence chains for q . A chain is sufficient if it supports a^* under the task definition and minimal if removing any required unit destroys that sufficiency. The set-valued definition accommodates alternative valid sources and paths.

A system exposes the following observable sequence:

$$L_N = \mathcal{R}_N(q, \mathcal{K}), \quad (1)$$

$$E_K = S_K(q, L_N), \quad E_K \subseteq L_N, \quad (2)$$

$$\pi(E_K) = O(q, E_K), \quad (3)$$

$$\hat{a} = \mathcal{G}(q, \pi(E_K)), \quad (4)$$

where L_N is the retrieval pool, E_K is a selected context under token or passage budget K , π is the reader-facing ordering or packing policy, and $\hat{a}$ is the generated answer. We use K as a generic context budget rather than only a passage count because compression and heterogeneous evidence make unit lengths variable.

3.3 Running Example

Consider the HotpotQA question: “What government position was held by the woman who portrayed Corliss Archer in the film *Kiss and Tell*?” The first support unit identifies Shirley Temple as the actor who portrayed Corliss Archer; the second states that she later served as United States Chief of Protocol [161]. The second unit is poorly specified by the original query and becomes straightforward to retrieve only after the bridge entity has been resolved.

This example separates five failures. The biography may be missing from L_N ; it may appear in the pool but be removed from E_K ; both passages may be selected but placed or compressed so that the reader cannot connect them; the reader may fail to compose the cast and biographical facts; or it may produce the correct title from parametric memory

without using the supplied chain. The same answer-level error can therefore originate at different stages, and a correct answer does not certify retrieval-grounded reasoning.

3.4 Adjacent Problems and Scope Boundaries

Graph paths, question decompositions, and agent trajectories are possible representations of an evidence chain, but none is identical to the chain by definition. A graph system can retrieve a path that is not sufficient; an agent can execute several tool calls without acquiring dependent evidence; a chain-of-thought can be multi-step while remaining entirely parametric. This survey includes these systems only when their external evidence acquisition, selection, composition, or verification is central to the task.

Long-context systems also require care. Enlarging the input may reduce the need for repeated retrieval, but it does not remove dependency, budget, or position effects. Similarly, multimodal and table-text systems change the type of evidence unit while preserving the same acquisition and composition problem. These substrates are discussed where they reveal a distinct retrieval or fusion mechanism rather than as independent application domains.

4 A DIAGNOSTIC DECOMPOSITION OF GROUNDED SUCCESS

The evidence-chain view is used as an evaluation decomposition. It does not assume that components are independent, that the stages can be changed one at a time, or that a method has an intrinsic score independent of its corpus, budget, reader, and intervention protocol.

4.1 Five Stage Events

For a question q , define the following events:

$$\mathcal{P} = \{\exists C \in \mathfrak{C}^\star(q) : C \subseteq L_N\}, \quad (5)$$

$$\mathcal{S} = \{\exists C \in \mathfrak{C}^\star(q) : C \subseteq E_K\}, \quad (6)$$

$$\mathcal{O} = \{\exists C \in \mathfrak{C}^\star(q) : C \subseteq E_K, \text{access}_\pi(C) \leq r\}, \quad (7)$$

$$\mathcal{U} = \{\hat{a} = a^\star \text{ under the realized chain-containing, accessibility-controlled context}\}, \quad (8)$$

$$\mathcal{T} = \{\exists C \in \mathfrak{C}^\star(q) : C \subseteq E_K, \hat{a} = a^\star, \hat{a} \leftarrow C\}. \quad (9)$$

The accessibility function is protocol-specific. It can encode the latest support position, the maximum separation between dependent units, or the outcome of a controlled packing treatment. It must therefore be declared for the reader and context format being evaluated. The notation $\hat{a} \leftarrow C$ denotes the latent claim that the answer causally depends on a valid chain C ; it is not directly observed from answer text or citation overlap.

The grounded-success event is

$$Y_g = \mathcal{P} \cap \mathcal{S} \cap \mathcal{O} \cap \mathcal{U} \cap \mathcal{T}. \quad (10)$$

Applying the chain rule in pipeline order gives

$$P(Y_g) = P(\mathcal{P})P(\mathcal{S} \mid \mathcal{P})P(\mathcal{O} \mid \mathcal{P}, \mathcal{S}) \quad (11)$$

$$\times P(\mathcal{U} \mid \mathcal{P}, \mathcal{S}, \mathcal{O})P(\mathcal{T} \mid \mathcal{P}, \mathcal{S}, \mathcal{O}, \mathcal{U}). \quad (12)$$

Equation 12 is an exact bookkeeping identity once the population and evaluation protocol are fixed. It is not a product model for predicting end-to-end performance. Its practical role is to specify the conditional subset on which each claim

Table 3. Evidence-chain stages, suitable diagnostics, and common confounds.

Stage	Target quantity	Suitable diagnostic	Main confound
Pool construction	$P(\mathcal{P})$	complete-chain or path recall at Top- N	incomplete or non-unique gold chains
Budgeted selection	$P(\mathcal{S} \mid \mathcal{P})$	chain preservation under a fixed token or passage budget	changing retrieval depth or evidence granularity
Presentation	$P(\mathcal{O} \mid \mathcal{P}, \mathcal{S})$	fixed-membership ordering or packing ablation	changing membership and order together
Fusion	$P(\mathcal{U} \mid \mathcal{P}, \mathcal{S}, \mathcal{O})$	oracle-evidence or controlled-context EM/F1; step validity	parametric answer availability
Causal faithfulness	$P(\mathcal{T} \mid \mathcal{P}, \mathcal{S}, \mathcal{O}, \mathcal{U})$	deletion, conflict, and bridge-counterfactual tests	redundant chains and out-of-distribution interventions

should be tested. Ordinary answer success $Y = \{\hat{a} = a^*\}$ also contains $Y \cap \neg Y_g$, including parametric-memory answers, single-hop shortcuts, incomplete but lucky reasoning, and guessing.

4.2 Conditional Evidence Contribution

Query–document relevance is an incomplete selection signal in a dependent retrieval task. For a fixed reader and partial context C , a useful mechanism quantity is the conditional contribution

$$u(e \mid q, C) = F_q(C \cup \{e\}) - F_q(C), \quad (13)$$

where $F_q(C)$ is the reader’s probability of the correct answer under context C . This signed, reader-dependent quantity is not a stage probability and is generally expensive to estimate. It nevertheless clarifies why a bridge passage can have low query-only relevance and high value after a core passage is known. Bridge-aware reranking, path scoring, note construction, and leave-one-out supervision estimate different surrogates of u .

4.3 Budget, Exposure, and Fusion

The transition from L_N to E_K is a constrained set-selection problem. A deeper pool can increase observability while adding distractors; a smaller reader context can reduce noise while discarding a complementary bridge. Compression changes both the cost and the semantic content of an evidence unit and should therefore be evaluated for chain preservation and provenance, not only token reduction.

Presentation must be separated from membership. If a reranker changes both which passages are selected and where they appear, an answer gain cannot be attributed to exposure. The cleanest test fixes E_K and varies only $\pi(E_K)$. Fusion should likewise be evaluated on examples where a valid chain is supplied in a controlled context. Oracle-evidence accuracy is not a deployment score; it is a reader diagnostic for bridge resolution, comparison, aggregation, or verification once evidence acquisition has been removed as a failure source.

4.4 Identifiability under Partial Annotation

Supporting-fact and path annotations identify only annotated versions of the pool, selection, and exposure events. A positive match establishes that at least one labeled chain is present; a negative match does not rule out an unannotated valid chain. Fusion and causal faithfulness are less identifiable from observational outputs. A correct answer with

Table 4. Design choices for evidence acquisition. “Diagnostic” denotes the minimum evidence needed to support an observability claim.

Family	State for the next query	Search unit / budget	Diagnostic	Characteristic risk
Single-shot sparse or dense	original question	passages; fixed Top- N	complete-chain recall at fixed N	later hops absent from the query surface
Iterative / interleaved	question plus retrieved or generated state	passages; hop or call budget	chain recall by hop and call count	query drift and error propagation
Graph expansion / traversal	local entities, paths, or sub-graph state	nodes, edges, linked text; expansion depth	path or chain recall plus graph coverage	missing links, noisy linking, explosive neighborhoods
Hierarchical / coarse retrieval	summaries, clusters, or hypothetical units	variable-granularity units; token budget	chain coverage at matched reader tokens	compression and granularity confounds
Agentic / RL search	reasoning history, tool observations, memory	queries and actions; learned stopping	chain recall, actions, tokens, and latency	more compute and weak action attribution

a plausible citation can still be produced from parametric memory. Evidence removal, factual conflict, and bridge substitution provide stronger evidence, but deletion can create unnatural inputs and alternative valid chains can mask sensitivity. Stage metrics should therefore be reported as protocol-bound estimators with their annotation and intervention assumptions made explicit.

5 METHOD FAMILIES ALONG THE EVIDENCE-CHAIN PIPELINE

We code a method by the earliest evidence-chain stage directly changed by its central mechanism and record later effects as spillovers. This rule prevents an architectural label from standing in for a causal claim: graph methods can alter acquisition, selection, or fusion; decomposition can generate retrieval queries or simplify reader computation; and an agent can act across the full pipeline. Each subsection therefore asks both what state the method uses and what evidence would be required to validate its stage-aligned claim.

5.1 Evidence Acquisition and Chain Observability

Acquisition methods differ mainly in the state used to issue the next query, the unit expanded, and the rule that stops search. Table 4 summarizes these distinctions.

Single-shot pool construction. BM25 [107], learned sparse retrieval such as SPLADE [25], dense retrievers including DPR [56], REALM [36], ANCE [152], CONTRIEVER [45], GTR [94], E5 [136], BGE [150], and RETROMAE [149], and late-interaction systems such as ColBERT and ColBERTv2 [57, 108] provide scalable candidate pools. They are indispensable baselines, but their query state is fixed. When a later hop is specified only by a bridge revealed inside earlier evidence, changing the matching function alone does not remove the structural information gap.

Iterative and reasoning-interleaved retrieval. GOLDEN RETRIEVER [89], MDR [153], BALEEN [58], and MUPPET [23] condition later retrieval on accumulated evidence or bridge state. IRCOT [134] uses intermediate reasoning as a query state; FLARE [53], ITER-RETGEN [114], ADAPTIVE-RAG [47], DRAGIN [119], SELF-RAG [4], and ITER-DRAGON [146] adapt retrieval timing, depth, or filtering to generator state. Their common benefit is state-conditioned reachability; their common confound is that more calls also change compute, pool size, and downstream context.

Graph and hierarchical acquisition. Question-specific graph systems such as GRAFT-NET [120], PULLNET [121], HGN [22], and DFGN [106] expand through entity–passage structure. ToG, ToG 2.0, and RoG [77, 81, 122] traverse knowledge-graph paths under language-model control, while KG-FiD and GNN-RAG [87, 168] use graph signals to prune or aggregate retrieved evidence. Corpus-level graph indexes such as GRAPH-RAG, LIGHT-RAG, and HIPPO-RAG [2, 20, 34]

Table 5. Selection mechanisms under a fixed reader budget.

Mechanism	Decision unit	Conditioning signal	What it can preserve	Main confound
Pointwise reranking	one document	query–document relevance	individually relevant supports	low-surface bridge evidence
Listwise reranking	ordered candidate list	query plus candidate interactions	relative membership and order	membership and exposure changed together
Bridge / path selection	next node, passage, or path	partial chain or bridge entity	complementary chain elements	dependence on noisy intermediate state
Plan / subquestion selection	evidence per subtask	explicit decomposition or plan	step-aligned evidence	decomposition errors and duplicated evidence
Compression / notes	tokens, spans, or generated notes	salience or reader contribution proxy	more evidence within token budget	lost relation, provenance, or factual detail

change the retrieval substrate itself. Hierarchical methods including RAPTOR, LONGRAG, and HIRQA [13, 54, 109] change evidence granularity, while HyDE [26] changes the query representation through a hypothetical document. For these systems, path reachability or summary coverage should be reported separately from the reader gain produced by a coarser context.

Agentic and adaptive search. REACT and TOOLFORMER [110, 164] made search and tool invocation part of the reasoning trajectory. Recent systems span reinforcement-learned search agents—including SEARCH-R1 [55], R1-SEARCHER [118], RESEARCH [9], and RAG-GYM [151]—and adaptive or structured controllers such as DEEP-RAG [33], PAR-RAG [170], DUAL-RAG [14], SEARCH-O1 [70], and MULTICUBE-RAG [115]. These systems place query generation, retrieve-or-reason decisions, review, document analysis, or structured expansion inside an adaptive control loop, but their training regimes differ substantially. Answer accuracy alone cannot show whether a controller improved chain observability or merely spent more reasoning and retrieval budget. At minimum, comparisons should match or expose retrieval calls, tokens, stopping behavior, and complete-chain recall.

5.2 Budgeted Evidence Selection and Context Construction

Once a chain is present in L_N , selection determines whether it survives in E_K . The key distinction is the decision unit: an independently scored document, a list, a path, a set conditioned on partial evidence, or a compressed representation. Table 5 summarizes the consequences.

Cross-encoders and neural rerankers, from passage reranking and MONOT5 to RANKT5, RANKLLAMA, RANKGPT, and RANKZEPHYR [79, 96, 97, 102, 123, 180], primarily estimate query-conditioned relevance. Listwise language-model rerankers [129] can model candidate interactions, but a better listwise score is not yet evidence that a complementary bridge was preserved. The required diagnostic is complete-chain preservation under the same pool and final token budget.

Bridge- and path-aware methods explicitly condition on partial state. BALEEN [58], HOPRETRIEVER [68], PATHRETRIEVER [3], BEAMDR [173], MUGER2 [162], and BRIDGE-R [156] score next-hop candidates, bridge entities, or chain hypotheses using more than the original query. Their design is closer to the conditional contribution $u(e \mid q, C)$, but the score is reader- and state-dependent and should not be presented as a universal notion of relevance.

Decomposition and planning methods such as DECOMPRC, SELF-ASK, LEAST-TO-MOST, BREAK, PLAN-AND-SOLVE, and PROGRAMFC [90, 98, 103, 137, 145, 176] convert a complex information need into subqueries or executable steps. This can make evidence selection more nearly single-hop, but it also introduces a new failure source: a wrong or overly coarse decomposition can make the valid chain unreachable before ranking begins.

Compression methods including LLMINGUA and LONGLLMLINGUA-2 [50, 100], RECOMP [154], CHAIN-OF-NOTE [169], and xRAG [15] alter both the budget and the evidence representation. A compressed context may retain an

answer-bearing phrase while removing the bridge relation or source provenance. Compression should therefore report chain preservation and citation validity at matched tokens, not only compression ratio and answer accuracy.

Supervision for selection can come from gold supports, path annotations, reader-derived leave-one-out labels, verifier scores, or synthetic counterfactual chains. Each signal has bias: gold supports are incomplete, reader-derived contribution is model-specific, and verifier scores may reward plausibility. A strong study states the surrogate and reports whether its gain survives a reader change.

5.3 Evidence Presentation and Exposure

Exposure concerns how a fixed evidence set is made available to a particular reader. Ordering and packing heuristics respond to documented position sensitivity [5, 75, 166]; learned approaches such as PRCA [160] and position-robust training [39] alter the layout or reader. Fusion-in- decoder architectures such as FiD, FiD-LIGHT, KG-FiD, and ATLAS [1, 44, 46, 168] reduce interference caused by concatenating all passages into one encoder sequence, but they do not remove selection failure. Compression can also improve access by shortening distractors, while simultaneously changing membership at the token level.

The decisive experiment is a fixed-membership treatment: hold E_K , the reader, and the total budget constant, then vary only ordering, grouping, or packing. Without that control, an “ordering” gain may be a hidden selection gain. Exposure is reader-specific, so the same layout should be tested across at least the reader families for which a general claim is made.

5.4 Cross-Evidence Fusion

Fusion methods act after usable evidence has reached the reader. Retrieve- then-read systems such as DRQA, ORQA, RAG, REALM, ATLAS, and FiD [8, 36, 44, 46, 62, 63] differ in retrieval training and passage encoding, but their fusion claim is best isolated with oracle or controlled evidence. Graph-aware readers such as QA-GNN, GREASELM, UNIKGQA, REAREV, and NSM [38, 51, 86, 165, 171] constrain composition through explicit structure.

Decomposition and programmatic reasoning reduce the complexity of the reader operation. DECOMPRC, SELF-ASK, LEAST-TO-MOST, DECOMPOSED PROMPTING, SUCCESSIVE PROMPTING, and REASONPATH [19, 59, 69, 90, 103, 176] externalize intermediate steps; PoT, PAL, FAITHFUL CoT, and code-based reasoning [12, 27, 78, 142] make some operations executable. Sampling and search over reasoning traces, including self-consistency, Tree-of-Thought, and Graph-of-Thought [6, 141, 163], can increase the chance that one trajectory composes the evidence correctly, but also add inference cost.

A fusion claim requires a chain-containing context and a declared presentation policy. If accuracy remains low under oracle evidence, the reader or decomposition is the binding failure. If accuracy improves only when the retrieval pool changes, the result should not be attributed to fusion alone.

5.5 Verification and Causal Faithfulness

Verification methods differ in where their evidence comes from. Self- verification and critique methods such as SELF-REFINE, REFLEXION, CoVe, SELF-CHECKGPT, and self- verification prompting [18, 82, 84, 116, 144] reuse the model’s own judgments. Externally grounded approaches including CRITIC, RARR, VERIFY-AND-EDIT, CRAG, SELF-CHECKER, and attribution-oriented traces [28, 31, 65, 71, 158, 175] retrieve or check evidence before revision. Process-supervised systems such as the process reward model of Lightman et al., MATH-SHEPHERD, REARTER, RAG-GYM, HiPRAG, and VERITAS [73, 124, 140, 148, 151, 157] move step-level reasoning, search-decision, or faithfulness signals into training.

Table 6. Representative recent systems and the diagnostics needed to support their coded stage-aligned interpretation. The table describes mechanisms, not a comparative performance ranking.

System	Central control signal	Primary target (secondary effect)	Required supporting diagnostic
Search-R1 [55]	RL-trained query generation during reasoning	observability	complete-chain recall and cost at matched calls
DeepRAG [33]	retrieve-or-reason policy	observability (cost control)	retrieval frequency, chain coverage, and accuracy-cost curve
RAG-Gym [151]	process supervision for search actions	observability (selection)	step-level rewards plus pool, context, and cost diagnostics
PAR-RAG [170]	top-down plan-act-review loop	observability (fusion)	plan validity, chain recall, and review ablation
VERITAS [157]	fine-grained trace-faithfulness rewards	causal faithfulness	information-think, think-answer, and think-search faithfulness
MultiCube-RAG [115]	ontology-guided structured retrieval	observability	chain coverage, noise, and ontology-construction cost

Their supervision targets differ: HiPRAG penalizes suboptimal search decisions, whereas VERITAS rewards trace-level faithfulness among information, reasoning, answers, and search actions.

Citation-generating systems such as WEBGPT, GOPHERCITE, and ALCE [29, 88, 92] make attribution inspectable. Deletion tests [74, 155] and bridge counterfactuals such as CoFCA [147] intervene on evidence more directly. These mechanisms should not be collapsed into one “faithfulness” score: self-critique tests consistency, citation metrics test entailment and coverage, and counterfactuals provide stronger but still imperfect evidence of causal use.

5.6 Hybrid and Agentic Systems: The Attribution Problem

Hybrid systems combine search, selection, reasoning, and verification, so an architectural name is least informative precisely where the system is most capable. Table 6 illustrates how recent systems can be described without treating the coded stage as a performance claim.

The general rule is simple: when an end-to-end system improves answer accuracy, the paper should show which observable object changed—the pool, the final context, its presentation, the reader outcome under controlled evidence, or the answer’s response to evidence interventions. Otherwise a gain from deeper search, more computation, or parametric answering is easily misattributed to the advertised component.

5.7 Cross-Family Synthesis

Four cross-family findings follow from the diagnostic view. First, most architectures that advertise multi-hop capability still make their strongest claim at the acquisition boundary: iterative retrievers, graph expanders, hierarchical indexes, and search agents all try to make an otherwise latent chain reachable. Their results are only comparable when retrieval depth, call count, and pool size are reported together.

Second, budgeted selection remains the least cleanly isolated middle stage. Rerankers, decomposers, compressors, and graph pruners often change both membership and ordering, so answer gains cannot by themselves show that a valid chain survived the final budget. This is where complete-chain preservation under matched passage or token budgets is most valuable.

Third, long context and graph structure help only when exposure is measured separately from membership. A larger context, community summary, or graph path can increase the chance that evidence is present while still placing dependent units too far apart, overcompressing the bridge, or obscuring provenance.

Fourth, causal faithfulness remains under-instrumented relative to answer accuracy and citation quality. Verification and attribution methods make support more inspectable, but deletion, conflict, and bridge-counterfactual probes are needed before a paper can claim that the supplied chain caused the answer rather than merely agreeing with it.

6 ARCHITECTURAL CROSSWALK

The diagnostic organization does not replace architecture. It clarifies why architectural labels alone are insufficient for comparison. Table 7 maps common families to the stages they most often affect. The mapping is non-exclusive: an iterative retriever can also perform selection, a graph system can support both acquisition and fusion, and an agent can act across the full pipeline.

Table 7. Architectural families and their typical evidence-chain effects. A filled circle marks a common primary effect and an open circle a frequent secondary effect.

Family	Observability	Selection	Exposure	Fusion	Faithfulness
Single-shot retrieval	•	◦			
Iterative / adaptive retrieval	•	•	◦		
Graph-guided retrieval	•	•		◦	◦
Decomposition / planning	◦	•		•	◦
Compression / context packing		◦	•	◦	
Fusion readers / structured reasoning			◦	•	
Verification / attribution				◦	•
Agentic / RL search	•	•	◦	◦	◦

Three distinctions are especially important. First, graph-based methods are not a single stage: graph construction can increase candidate coverage, subgraph pruning can change selection, and message passing can change fusion. Second, “long context” is not a substitute for retrieval analysis. It changes the feasible budget and evidence granularity but can preserve or worsen position sensitivity. Third, agentic systems add adaptive control rather than a new diagnostic target. Their main evaluation problem is attribution: additional calls can improve answer accuracy through deeper search, better selection, more computation, or self-correction. Per-stage and cost-normalized results are therefore more informative than a single agent-versus-pipeline comparison.

7 BENCHMARKS AS DIAGNOSTIC INSTRUMENTS

A benchmark is diagnostically useful when its annotations or interventions make a failure stage observable while controlling preceding stages. Domain and release year matter, but they are secondary to what the benchmark can identify.

7.1 Supporting Facts, Paths, and Chain Coverage

HOTPOTQA, 2WIKIMULTIHOPQA, IIRC, COMPLEXWEBQUESTIONS, and QANGAROO [24, 41, 127, 143, 161] provide varying degrees of support or path information. These labels allow pool-level and context-level coverage diagnostics, but they identify annotated chains rather than every valid chain or the chain actually used by the model. 2WIKIMULTIHOPQA, ENTAILMENTBANK, EQASC, and PROOFWRITER [17, 41, 48, 126] make more intermediate structure explicit and are therefore better suited to path or step analysis. Even then, a gold path need not be unique.

7.2 Shortcut Resistance and Longer Dependencies

MUSIQUE [133] was designed to reduce single-hop shortcuts in composed questions. MOREHOPQA [112] stresses longer chains, while HOTPOTQA-NO SHORTCUT [132] and related filtered settings reduce examples that can be solved from one support. NOVELHOPQA [35] combines dependency with longer narrative context. These datasets are useful for acquisition claims, but a low end-to-end score still cannot reveal whether the chain was missing, removed by the context budget, or mishandled by the reader.

7.3 Heterogeneous Evidence and Long Context

HYBRIDQA, OTT-QA, MULTIMODALQA, WEBQA, MANYMODALQA, TAT-QA, FETAQA, VISDOM, DOCHOP-QA, and WIKIMIXQA [7, 10, 11, 37, 40, 93, 125, 128, 139, 179] require chains that cross text, tables, documents, or images. Their primary value is to test whether evidence representation and fusion remain coherent across substrates; alignment errors can dominate ordinary retrieval error.

Long-context suites such as LONGBENCH, ∞ BENCH, HELMET, LOONG, and ZEROSCROLLS [5, 113, 138, 167, 172] can reveal access failures, but their default protocols rarely isolate presentation from membership. To support an exposure claim, a study should construct fixed-set variants that move the same support chain across positions or separations while holding tokens and reader constant.

7.4 Support, Attribution, and Counterfactual Use

Fact-verification datasets such as FEVER, HOVER, EX-FEVER, SCIFACT, and AVERITEC [52, 80, 111, 131, 135] measure whether claims are supported or refuted by evidence. Citation-oriented benchmarks such as ALCE [29], together with HALUEVAL and EXPERTQA [67, 83], measure attribution, factuality, or support quality. These are important faithfulness proxies, but support entailment does not show that the evidence caused the answer. Bridge-counterfactual protocols such as COFCA [147], conflict tests, and carefully controlled deletion experiments provide stronger evidence because they intervene on the chain.

7.5 End-to-End RAG and Agentic Behavior

MULTIHOP-RAG, FRAMES, CRAG, FANOUTQA, and RAGTRUTH [60, 95, 130, 159, 178] measure end-to-end RAG behavior under complex information needs. TOOLBENCH, AGENTBENCH, and WEBARENA [76, 105, 177] add tool or environment interaction. They are valuable deployment tests, but their scores combine action policy, retrieval affordances, context construction, reader reasoning, and cost. Stage-level logs are required for mechanism claims.

Table 8. Benchmark families interpreted as diagnostic instruments.

Family	Available signal	Best-identified target	Principal blind spot
Supporting-fact QA	document or sentence supports	observability and selection	incomplete chains and shortcut answers
Path / proof annotated	ordered paths, trees, or steps	fusion and process validity	alternative valid derivations
Shortcut-resistant QA	composed or filtered questions	dependent acquisition	stage of failure remains mixed
Heterogeneous evidence	table, text, graph, or image links	cross-substrate fusion	alignment and representation confounds
Long-context suites	long inputs and position variation	exposure when membership is controlled	default tasks mix selection and layout
Verification / citation	support labels, entailment, citations	support validity and attribution	causal use is not identified
Counterfactual / conflict	altered bridge or evidence facts	causal faithfulness	intervention naturalness and redundancy
Agentic / tool use	action traces and environment outcomes	joint control and efficiency	tool affordances dominate comparisons

Table 9. Metrics and the claims they can support when the required control is satisfied.

Claim	Suitable metric	Conditional subset / control	Misleading substitute
Pool contains a valid chain	complete support or path recall at Top- N	fixed corpus, evidence unit, and retrieval budget	answer F1 alone
Budget preserves the chain	selected-chain recall at K	chain present in pool; fixed token budget	reranker relevance score
Presentation improves access	fixed-set ordering or packing gap	identical membership and reader chain supplied and presentation declared	reranking that also changes membership
Reader composes evidence	oracle- or controlled-evidence EM/F1; step validity	answer-correct and chain-present subsets	Top- N retrieval recall
Answer is faithfully supported	joint citation, deletion, conflict, and counterfactual diagnostics	matched model/API/hardware conditions	citation overlap alone
System is efficient	tokens, calls, latency, cost-quality curve		selector-only latency

7.6 Metric–Claim Alignment

Answer EM and F1 are deployment-level metrics rather than stage-aligned metrics. Supporting-fact F1 approximates annotated membership; Top- N recall estimates pool observability; citation precision estimates attribution; and counterfactual sensitivity estimates response to an intervention. None is a universal “multi-hop RAG quality” score.

A balanced empirical portfolio should include at least one benchmark with chain annotations, one that reduces shallow shortcuts or increases dependency length, and one that tests support or counterfactual behavior. No individual dataset substitutes for conditional evaluation on the system’s own pool, context, ordering, reader, and evidence interventions.

8 FAITHFULNESS AS CAUSAL EVIDENCE USE

Correctness, attribution, and causal use are distinct. A model can answer correctly without citing evidence, cite an entailing passage that it did not use, or produce a well-supported chain whose final composition is wrong. Faithfulness evaluation should therefore ask how the answer responds when the relevant evidence is withheld, contradicted, or altered, rather than treating a plausible citation as proof of use.

8.1 Operational Probes

For a candidate chain $C \subseteq E_K$, a simple deletion statistic is

$$D_{\text{del}}(C) = P(a^\star \mid q, \pi(E_K)) - P(a^\star \mid q, \pi(E_K \setminus C)). \quad (14)$$

A large value suggests that C is load-bearing, but it is not a definition of faithfulness. Deletion can create an unnatural input, and redundant valid chains can make a faithful answer insensitive to any one removal.

Counterfactual bridge substitution is often stronger than deletion because it preserves a coherent context while changing the fact that should propagate through the chain. Closed-book and evidence-conflict controls estimate the

Table 10. Complementary probes of evidence use. No single row identifies causal faithfulness in all settings.

Probe	What it asks	Strong evidence when	Failure mode
Closed-book / conflict control	does the answer follow supplied evidence rather than parametric memory?	evidence conflicts are coherent and answerable	model may distrust artificial conflicts
Deletion / sufficiency	does removing a support chain reduce the answer probability?	input remains natural and alternatives are controlled	redundancy and distribution shift
Bridge counterfactual	does an altered intermediate fact propagate to the answer?	replacement yields a coherent alternative chain	expensive construction and leakage from unchanged cues
Citation entailment / coverage	do cited units support the generated claims?	claim segmentation and entailment judgments are reliable	post-hoc citation without causal use

availability and dominance of parametric knowledge. Citation entailment is necessary for auditable output, but it measures support compatibility rather than the internal cause of generation. These probes should be combined rather than averaged into an opaque single score.

8.2 Reporting Faithfulness

A credible audit reports at least three joint quantities on the answer-correct subset: citation entailment or support precision, sensitivity to a declared evidence intervention, and closed-book or conflict behavior. Results should also be stratified by whether an annotated chain is present in the final context. Faithfulness on chain-absent examples mainly measures parametric answering or retrieval failure; faithfulness averaged over all examples can hide that distinction.

Verification methods should be described by the stage at which they act. Self-critique tests internal consistency, externally grounded verification checks or retrieves additional support, process supervision changes the training objective, and counterfactual training changes the data distribution. Their outputs are complementary evidence about causal use, not interchangeable measurements of one universal property.

9 OPEN PROBLEMS AND TRADEOFFS

Dependent retrieval under controlled cost. Iterative and agentic search can expose bridge evidence that is invisible to the original query, but each step also increases latency, distractor mass, and query-drift risk. Progress requires accuracy–cost curves, chain-recall curves by call count, and stopping policies trained against evidence acquisition rather than only final-answer reward.

Set- and path-level selection. Most rerankers still score documents independently or listwise against the original query. Multi-hop selection requires complementarity: a passage can be weak alone and decisive with another passage. Core-conditioned scoring, path-level objectives, set selection, and counterfactual contribution labels are promising, but they must be evaluated under fixed pools and reader token budgets.

Evidence granularity, compression, and provenance. Passages, sentences, graph triples, summaries, and generated notes induce different chain lengths and costs. Compression can preserve an answer string while removing the relation or provenance required to justify it. Studies should report the evidence unit, token accounting, chain preservation, and citation recoverability after compression.

Reader-specific context organization. Long-context models do not make presentation irrelevant. Ordering, separation, grouping, and instruction layout interact with the reader architecture and training distribution. Exposure metrics should therefore be reader-specific, and presentation claims require fixed-membership tests across the reader families to which the claim is intended to generalize.

Table 11. Minimum evidence-chain reporting for multi-hop RAG studies.

Reported quantity	Claim	Required control
Chain/path recall at Top- N	pool observability	fixed corpus, retrieval budget, and evidence unit
Chain preservation at budget K	selection	chain present in the pool; fixed token budget
Order/packing gap on a fixed set	exposure	identical evidence membership, budget, and reader
Controlled-context EM/F1	fusion	valid chain supplied and presentation specified
Entailment plus intervention probes	causal faithfulness	answer-correct and chain-present subsets
Tokens, calls, and end-to-end latency	efficiency	comparable model, API, and hardware conditions

Identifiable benchmarks with alternative chains. Many benchmarks annotate one support path even when several valid chains exist. New evaluation instruments should include alternative chain labels, controlled distractor density, fixed-set ordering variants, oracle contexts, and coherent bridge counterfactuals. These additions are more informative than another end-to-end leaderboard with no failure localization.

Attribution and robustness in adaptive systems. Agents, graph builders, and verifiers increase both capability and attack surface. Retrieval-side prompt injection, poisoned or stale corpora, source conflicts, and unverifiable generated graph edges can corrupt the chain while leaving the answer fluent. Trust-aware retrieval, provenance-preserving context construction, and per-action audit logs should be evaluated alongside accuracy and efficiency.

10 REPORTING GUIDELINES FOR MULTI-HOP RAG

The evidence-chain framework implies a compact reporting protocol. A paper need not optimize every stage, but it should measure the stage it claims to improve, expose major spillovers, and state the budget under which the claim holds.

Four practices are necessary for interpretable comparison. First, retrieval depth N , final context budget K , evidence granularity, reader model, and maximum calls must appear next to the headline score. Second, stage-aligned metrics must be reported on the corresponding conditional subset: selection on chain-present pools, exposure on fixed membership, fusion on controlled evidence, and faithfulness on answer-correct cases with available support. Third, hybrid and agentic systems need stage-aligned ablations and cost-quality curves because additional computation can conceal the source of an improvement. Fourth, benchmark and survey papers should publish their coding assumptions and distinguish verified records from discovery records.

Incomplete gold chains make recall a lower-bound proxy; reader-derived contribution is model-specific; deletion sensitivity is affected by redundancy; and citation entailment does not establish causal use. Stating these limitations produces more comparable evidence than assigning one overloaded score to “RAG quality.”

11 CONCLUSION

Multi-hop RAG is an information-retrieval problem over dependent evidence, not simply a larger retrieve-then-read pipeline. The evidence-chain view separates five questions that end-to-end answer accuracy conflates: whether a valid chain enters the pool, survives the reader budget, is organized for access, is composed correctly, and faithfully supports the answer. This organization connects sparse and dense retrieval, graph expansion, decomposition, compression, fusion readers, verification, and agentic search without treating an architectural family as an explanation of performance.

The main methodological implication is conditional evaluation. Pool recall cannot support a reader claim; reranking that changes both membership and order cannot isolate exposure; oracle-evidence accuracy does not establish faithfulness; and citation overlap does not prove causal use. Each claim requires a corresponding observable object, subset, intervention, and budget. The companion artifact operationalizes the same discipline for the survey itself by separating discovery breadth from source-verified taxonomy records.

Future progress will depend less on adding another pipeline label and more on building controlled instruments for dependent acquisition, complementary selection, reader-specific context organization, cross-evidence composition, and counterfactual support. These instruments are what allow improvements to be compared across rapidly changing retrievers, readers, graph indexes, and search agents.

DATA AND ARTIFACT AVAILABILITY

The companion artifact is available at <https://github.com/TsingyuL/multi-hop-rag-survey>. It contains the search protocol, machine-readable method and benchmark records, coding fields, source URLs, status labels, figure sources, and validation scripts. Versioned releases allow the manuscript, bibliography, figures, and coded records to be associated with the same frozen snapshot.

A ARTIFACT STATUS AND UPDATE POLICY

The artifact separates three layers that serve different purposes. *Discovery records* maximize recall during literature search and may contain adjacent or unverified work. *Source-verified records* have been checked against the canonical paper or official project page and can support artifact-derived taxonomy analyses. *Manuscript citations* additionally include background and component papers needed for exposition. These layers overlap and should not be interpreted as a PRISMA-style screening funnel.

Each public release records its literature cut-off, canonical-source policy, status vocabulary, and correction history. Numerical catalog summaries are generated from the release rather than copied into the source by hand, which prevents stale counts and restricts artifact-derived tables to records that satisfy the stated verification rule.

B SCREENING AND CODING CHECKLIST

A method is promoted to reviewed status only after the following fields have been checked against the full paper or an official project page: canonical title and venue/version; multi-hop scope; evidence substrate; retrieval and reader budgets; architectural family; primary and secondary evidence-chain stages; benchmark and metric claims; source and code URLs; and a short passage supporting the taxonomy assignment. Comparative tables should be generated from reviewed records rather than maintained manually.

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